Markscheme

Mathematics: analysis and approaches
Practice question bank

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27. (a)

Consider the function $f(x) = \dfrac{8x^2 - 25x + 12}{(x-2)^2(x+1)}$.

Write $f(x)$ as the sum of partial fractions in the form $\dfrac{A}{x-2} + \dfrac{B}{(x-2)^2} + \dfrac{C}{x+1}$.

attempt to write $A(x-2)(x+1) + B(x+1) + C(x-2)^2$ *(M1)*

$8x^2 - 25x + 12 = A(x-2)(x+1) + B(x+1) + C(x-2)^2$ **A1**

substituting $x = 2$: $-6 = 3B \Rightarrow B = -2$ **A1**

substituting $x = -1$: $45 = 9C \Rightarrow C = 5$ **A1**

comparing x^2 coefficients: $8 = A + C \Rightarrow A = 3$ **A1**

[5 marks]

(b)

Hence find the exact value of $\displaystyle\int_3^4 f(x)\,dx$.

$$\int f(x)\,dx = 3\ln|x-2| + \frac{2}{x-2} + 5\ln|x+1| \qquad \textit{(M1)}$$

evaluating between $x = 3$ and $x = 4$ gives ≈ 2.195 **A1**

[2 marks]